

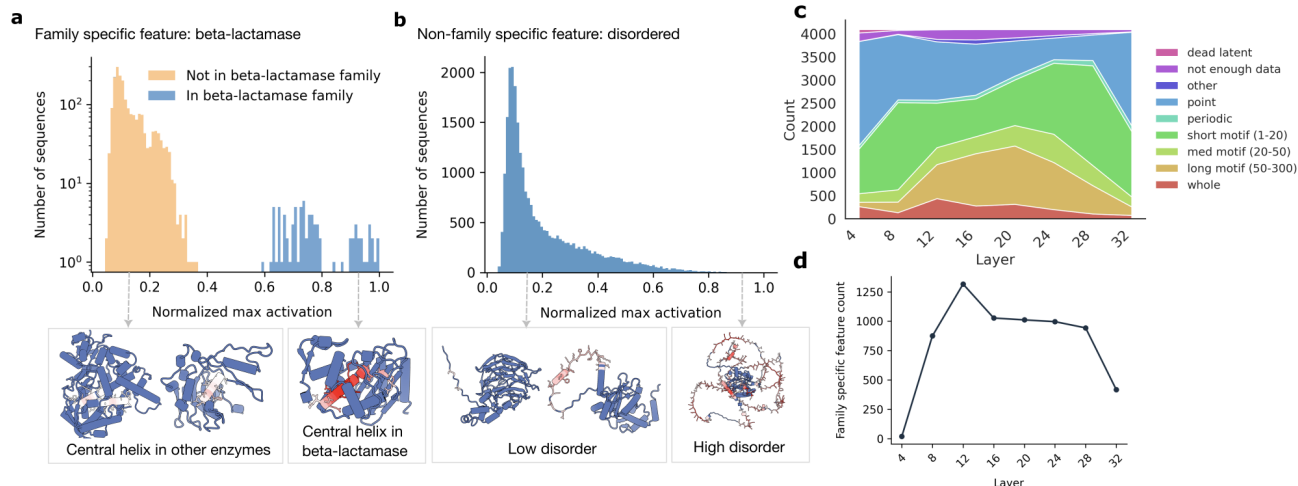


Figure 3. **a**. Example of a family specific feature, the central helix of a beta lactamase. The histogram shows the distribution of maximum activations in the sequences which activate the latent. **b**. Example of a non-family specific feature, corresponding to disordered sequences. **c**. Latents classified by activation pattern across layers. **d**. Number of family-specific latents across layers. The SAEs used in c and d have a hidden dimension of 4096 and $k=64$.

test sets based on a threshold of 40% sequence identity.

3.3. Linear probes

For each task and for varying ESM layers, we train linear probes on both ESM and SAE embeddings. We mean-pool embeddings over the length of the sequence for tasks with protein-level labels: subcellular localization, thermostability, and mammalian cell expression. For all linear regression and logistic regression tasks, we perform grid search over a range of regularization strengths and pick the best model using a validation set. See Table 1 for a summary of the probe implementation for each task.

3.4. Manual feature interpretability study

Following Rajamanoharan et al. (2024), we conducted a blinded study in which seven participants (graduate students and undergraduates familiar with protein biology) rated the interpretability of SAE and ESM latents from layer 24. Embeddings were computed on sequences from Swiss-Prot clustered at 30% sequence identity (see subsection A.6). Using our InterProt visualizer, each participant viewed 100 randomly selected features (drawn uniformly from both models) and rated their interpretability as *yes*, *no*, or *maybe*.

4. Feature Analysis

4.1. SAEs uncover interpretable features

We train TopK SAEs on the residual stream of different layers of ESM-2 (650M) to explore its learned representations.

To interpret a given SAE latent dimension (*i.e.*, a latent), we examine the top activating protein sequences, *i.e.*, those that produce the largest activation for that dimension. We developed a latent visualizer, *InterProt*, to streamline the process of identifying features. *InterProt* allows users to align the most-activating sequences and highlight their corresponding protein structure by activation strength. It is open source and can be accessed at <https://interprot.com>.

Through these visualizations, we gain insight into how ESM internally represents protein sequences (Figure 2). Many discovered features correspond to recognizable biological concepts, such as secondary structure elements (*e.g.*, alpha helices and beta strands), short conserved motifs, or entire functional domains. Some features capture biochemical concepts like cysteine-cysteine bonds or the orientation of residue side chains. In many ways, latents capture characteristics of a protein that a biologist might notice.

Notably, while many latents are intuitive (*e.g.*, focused on conserved motifs), others are unexpectedly specific. For instance, rather than a single, generic “alpha helix” latent, we observe multiple helix-related latents that activate under more context-dependent conditions (*e.g.*, at the beginning of a helix, only when the helix is buried, or when the helix meets a certain length criterion). Other examples include highly context specific amino acids, such as asparagine residues in disordered regions.

To systematically evaluate interpretability, we conducted a blinded human study to rate SAE and ESM latents (subsection 3.4). SAE latents were rated as interpretable far more

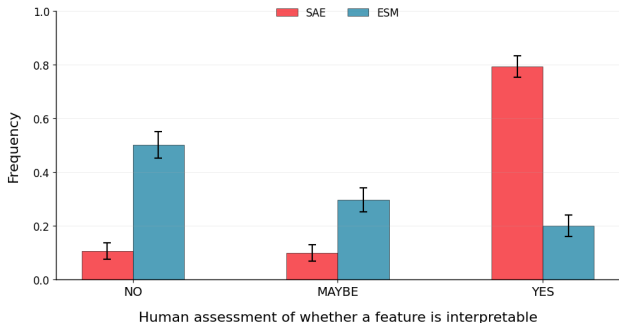


Figure 4. Human rater scores of SAE or ESM latents from layer 24. The error bars indicate the 95% confidence interval, assuming a binomial 1-vs-all distribution.

often than those from ESM (Figure 4). Approximately 80% of SAE features received a “yes” rating for interpretability, broadly consistent with previous findings on SAEs trained on language model activations (Rajamanoharan et al., 2024).

4.2. SAE latents reveal a learned notion of protein families

Intriguingly, a large subset of latents appear to be protein family-specific. These latents activate weakly on most sequences but strongly on proteins belonging to a particular family. For example, the latent shown in Figure 3a appears to highly activate only for the central helix in beta-lactamases, but only exhibits marginal activations when a similar structural motif appears in proteins outside the beta-lactamase family.

By contrast, a non-family-specific latent such as the “disordered” latent in Figure 3b, responds broadly to intrinsically disordered regions in proteins across many families. These observations collectively suggest that pLMs learn internal representations that, in part, mirror known sequence-homology based groupings of proteins, akin to a model-internal notion of protein families.

To better understand the relationship between a family-specific latent and its associated protein family, we investigate what happens when we “steer” ESM using the latent. Specifically, we set the activation of a family-specific latent to a multiple of its maximum activation, and continue the model’s forward pass using the reconstructed activation. We find that, compared to steering random latents which activate on a sequence, steering family specific latents causes fewer residues to change (Figure 7). Investigating how to steer the model in useful ways remains an open direction for future work.

4.3. Classifying latents

Motivated by these observations, we propose two classification schemes for the SAE latents.

Activation Pattern. Latents exhibit distinct activation patterns across their top sequences (Figure 1b). To capture these differences, we use a rule-based scheme that classifies features into categories such as: point (activates on single residues at a time), periodic (activates in a regular interval, repeating every n residues), motif (activates in short, medium, or long contiguous intervals), and domain (activates over nearly the entire sequence). Refer to Table 2 for precise definitions.

Family Specificity. To quantify family specificity, we use each latent for binary classification: given a protein, can this latent’s activations be used to predict membership in a particular family? We label a latent “family-specific” if it achieves an F1 score > 0.7 at a certain activation threshold. We use all protein sequences from Swiss-Prot (Boeckmann et al., 2003) (clustered as 30% sequence identity) and categorized into InterPro protein families (Paysan-Lafosse et al., 2023) for our evaluation. Refer to subsection A.6 for details.

These classifications allow us to systematically group SAE features and draw broader conclusions about their roles in representing protein structure and function.

4.4. Influence of SAE training choices on latent classification

Prior literature on LLMs suggests that SAE-learned features depend on multiple factors, including model hyperparameters (*e.g.*, the sparsity level and latent dimension size) and training data (*e.g.*, which layer’s activations are used). We therefore examine how these factors impact the types of features that emerge in pLMs.

Effect of k . In TopK SAEs, the k hyperparameter controls how many latents can be active for a given input, effectively setting the sparsity level. We find that, for a fixed hidden dimension, lower k (*i.e.*, higher sparsity) yields more family-specific features (Figure 8a). One plausible explanation is that family-specific features constitute the most salient signals used in reconstruction.

Effect of expansion factor. We also experiment with increasing the latent dimension, called the expansion factor. While we initially hypothesized that a larger latent dimension might shift the distribution of feature activation patterns, we observed that activation pattern classifications (point, motif, domain, etc.) remain largely consistent, while increasing the number of family specific features (Figure 8b).

Comparison across layers. Finally, we investigate layer-wise differences in the learned features. Previous studies have shown that downstream performance can vary substantially depending on which layer representations are used, suggesting that different layers capture different degrees of abstraction. In our experiments, family-specific features tend to peak in certain early-to-mid layers of ESM, then

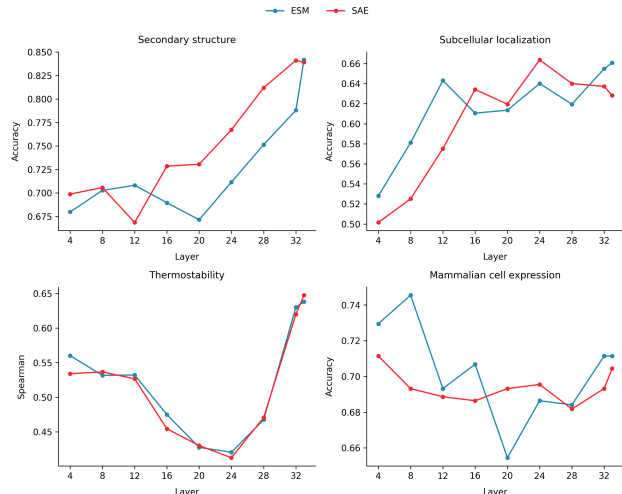


Figure 5. Linear probe performance on ESM vs. SAE embeddings across ESM layers. In all 4 downstream tasks, SAE performs competitively with ESM. For secondary structure prediction, SAE consistently outperforms ESM in most layers. All classifications tasks are measured using accuracy; thermostability is measured using Spearman’s rank correlation.

decline in later layers (Figure 3d). Similarly, features which activate contiguously over long sections (motif/domain) are more common in earlier layers, whereas later layers exhibit features with shorter contiguous activations (Figure 3c). Given that later layers are thought to specialize for final logit computation (Gao et al., 2024), one possibility is that shorter, more specific activations better serve this goal.

5. Interpretable Probing

Representations learned by pLMs can be used in downstream tasks. Often, this is as simple as training a linear model on the pLM embeddings. Although this leads to a predictive model, it does so at the expense of interpretability, as the coefficients of the linear model do not correspond to interpretable features. In this section, we explore the ability of SAE representations to overcome this limitation. Since SAE latents are often interpretable, those that contribute significantly to prediction could reveal biological reasons that explain the prediction.

5.1. SAE probes perform competitively with ESM probes

In the context of LLMs, linear probes on SAE embeddings can achieve performance similar to, or in some cases better than, linear probes on their base model embeddings (Bricken et al., 2024; Kantamneni et al., 2024). To explore this dynamic in pLMs, we train linear probes on ESM and SAE embeddings across different layers. Following Li et al. (2024), we evaluate the probes on benchmarked downstream tasks:

secondary structure, subcellular localization, thermostability, as well as a novel task, mammalian cell line expression prediction.

We find that linear probes on SAEs achieve performance similar to their ESM baselines across all layers (Figure 5). For secondary structure prediction, the SAE probe consistently outperforms the ESM probe, a result we discuss in the next section and attribute to the abundance of SAE latents that correspond to secondary structure.

5.2. SAE probes uncover interpretable latents corresponding to known mechanisms

To assess the interpretability of linear probes trained on SAE latents, we analyze the distribution of their coefficients (Figure 10) and manually inspect the latents with the largest coefficients in the InterProt visualizer¹. We find that those latents often correspond to biological concepts with known relevance to each task (Figure 6).²

5.2.1. SECONDARY STRUCTURE

Secondary structure prediction is a task that is known to utilize the features learned during pLM pre-training (Li et al., 2024). Our analysis of the highest-weighted SAE latents in our secondary structure probe confirms this and provides an explanation for why early layers perform worse.

Our SAE probes reveal that most layers contain features corresponding to secondary structure elements. The most positive coefficients for the alpha helix class are typically latents that activate exclusively on helices (Figure 6a). They range from specific helix residues, to entire helices, to helix-helix interactions. We observe similar results for the two other classes: beta strand and *other*, where *other* tends to correlate with features that activate on disordered regions.

Why is classification performance worse in early layers? We find that early layers do not contain generic features for secondary structure. For example, the top latent for helix classification in layer 4 (L4/2624) detects proline (P) residues, which often appear at helix boundaries due to their disruptive effect on hydrogen bonding. This reliance on a weak amino acid-level correlation highlights the absence of a more generalized helix feature. Similarly, the top latent for the disordered classifier activates on N-terminal residues, reflecting the spurious correlation that many proteins have disordered regions at their N-terminus, rather than capturing

¹Linear probe coefficients are not comparable across tasks because different tasks may apply different regularization strengths, optimized via grid search.

²In this section and in Figure 6, we discuss latents from a range of layers. The choice of layers is qualitative—primarily based on latents we find particularly interpretable or notable—and does not imply the lack of interesting latents in undiscussed layers.